

Heart Disease Analysis

Phase 3: Project Design Phase

Problem Solution Fit

Naga Nandini Padidham | [GitHub: nandinipadidham-igtm/Heart-Disease-Analysis](#)

Problem Statement

Heart disease is one of the leading causes of death worldwide, yet many of its risk factors — age, smoking, alcohol use, diabetes, poor sleep, and general health — are well documented but rarely brought together into a single, easy-to-explore view. Healthcare organizations and researchers often lack a fast, visual way to identify which patient groups are most at risk, making it harder to plan targeted screening and prevention programs.

Target Users

- Healthcare analysts who need to summarize risk patterns for reporting.
- Hospital administrators planning screening and outreach programs.
- Public health researchers studying lifestyle and demographic risk correlations.
- Students/aspiring data analysts learning end-to-end analytics workflows.

Problem–Solution Fit Table

Problem	How the Solution Addresses It
Risk factors are scattered across raw data with no visual summary.	A single interactive Tableau dashboard consolidates all key risk factors into one view.
Identifying high-risk age/lifestyle groups requires manual analysis.	SQL-driven aggregation + Tableau filters instantly surface high-risk segments (e.g. age 60-69, diabetics, smokers).
Static reports don't let stakeholders explore the data themselves.	Interactive filters (Gender, Smoking, Alcohol, Heart Disease) let users explore on their own terms.
Dashboards are often locked inside desktop tools, hard to share.	Dashboard is published to Tableau Public and embedded in a Flask web app for easy, shareable access.
Insights alone don't drive action without clear recommendations.	A guided Tableau Story ends with concrete, actionable recommendations for healthcare organizations.

Why This Matters

By closing the gap between raw healthcare data and actionable insight, this project demonstrates how a lightweight, low-cost analytics stack (MySQL + Tableau + Flask) can support meaningful public-health decision-making without requiring a large BI infrastructure investment.